DriverJinn — Downstream Analysis & Poster Plots

Setup

Training Curves

```
KEY = "val_ndcg@50"
fig, ax = plt.subplots(figsize=(8, 6))
for fold, history in sorted(histories.items()):
    if KEY not in history or len(history[KEY]) == 0:
        continue
    y = np.array(history[KEY])
    ax.plot(np.arange(1, len(y) + 1), y,
            label=f"Fold {fold}", color=FOLD_COLORS[fold - 1], alpha=0.85, linewidth=1.5)
ax.set_xlabel("Epoch", fontsize=12)
ax.set_ylabel("NDCG@50", fontsize=12)
ax.set_title("Validation NDCG@50 Across Folds", fontsize=14, fontweight="bold")
ax.legend(loc="best", fontsize=10)
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()
```

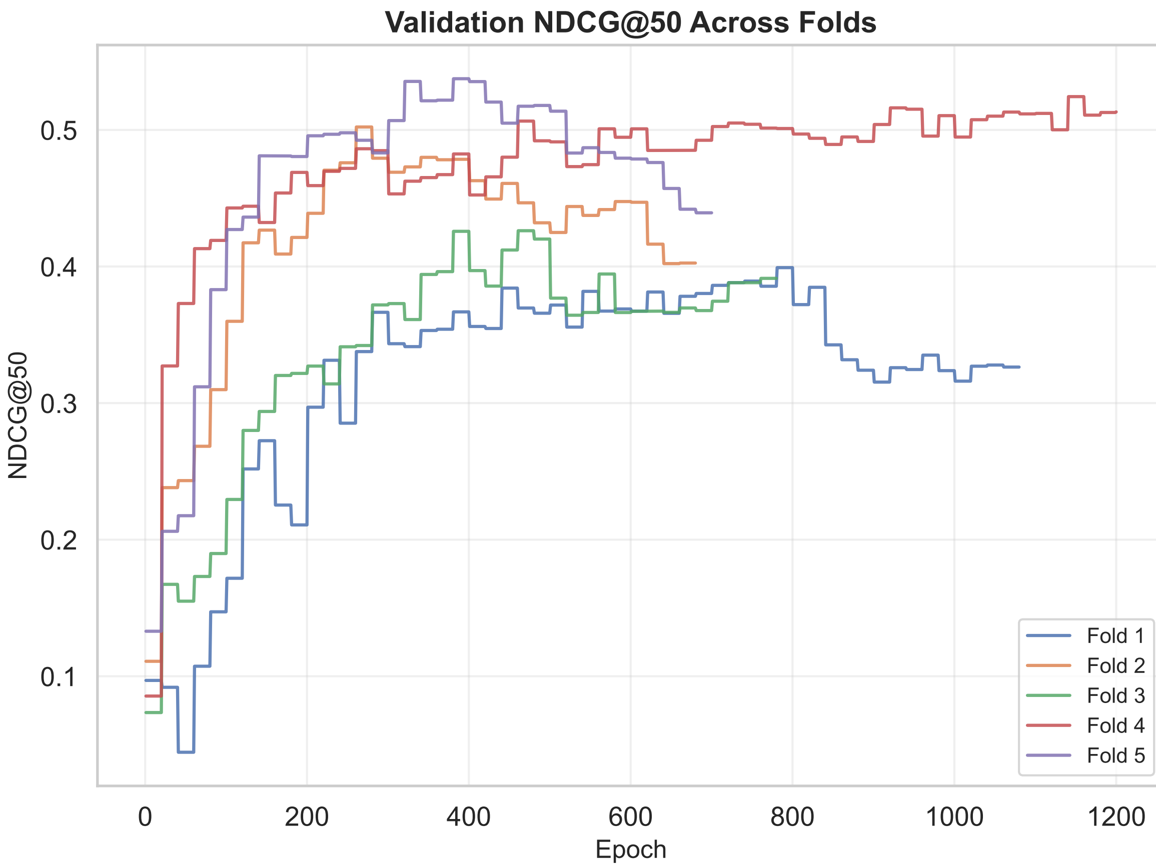


Figure 1: Validation NDCG@50 across folds.

```
KEY = "val_auroc"
fig, ax = plt.subplots(figsize=(8, 6))
for fold, history in sorted(histories.items()):
    if KEY not in history or len(history[KEY]) == 0:
        continue
    y = np.array(history[KEY])
    ax.plot(np.arange(1, len(y) + 1), y,
            label=f"Fold {fold}", color=FOLD_COLORS[fold - 1], alpha=0.85, linewidth=1.5)
ax.set_xlabel("Epoch", fontsize=12)
ax.set_ylabel("AUROC", fontsize=12)
ax.set_title("Validation AUROC Across Folds", fontsize=14, fontweight="bold")
ax.legend(loc="lower right", fontsize=10)
ax.yaxis.set_minor_locator(ticker.AutoMinorLocator())
ax.grid(True, alpha=0.3)
plt.tight_layout()
```

```
plt.show()
```

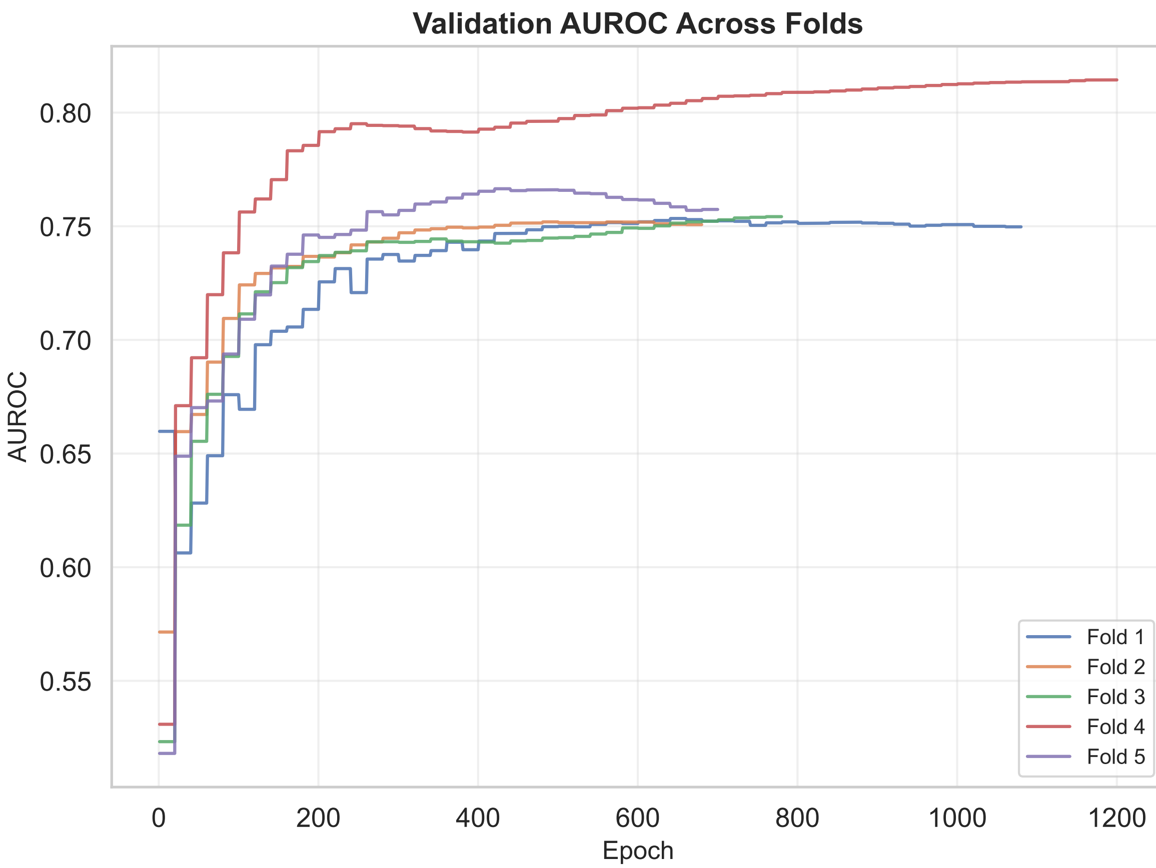


Figure 2: Validation AUROC across folds.

```
rows = []
for fold, history in sorted(histories.items()):
    ndcg  = history.get("val_ndcg@50", [])
    auroc = history.get("val_auroc", [])
    rows.append({
        "Fold":          fold,
        "Peak NDCG@50": round(max(ndcg),  4) if ndcg  else None,
        "Peak AUROC":   round(max(auroc), 4) if auroc else None,
        "Epochs":       len(ndcg),
    })
df_summary = pd.DataFrame(rows)
mean_row = {"Fold": "Mean",
```

```
            "Peak NDCG@50": round(df_summary["Peak NDCG@50"].mean(), 4),
            "Peak AUROC":   round(df_summary["Peak AUROC"].mean(),   4),
            "Epochs":       "-"}
df_summary = pd.concat([df_summary, pd.DataFrame([mean_row])], ignore_index=True)
df_summary.style.set_caption("Per-fold peak validation metrics")
```

Table 1: Per-fold peak validation metrics

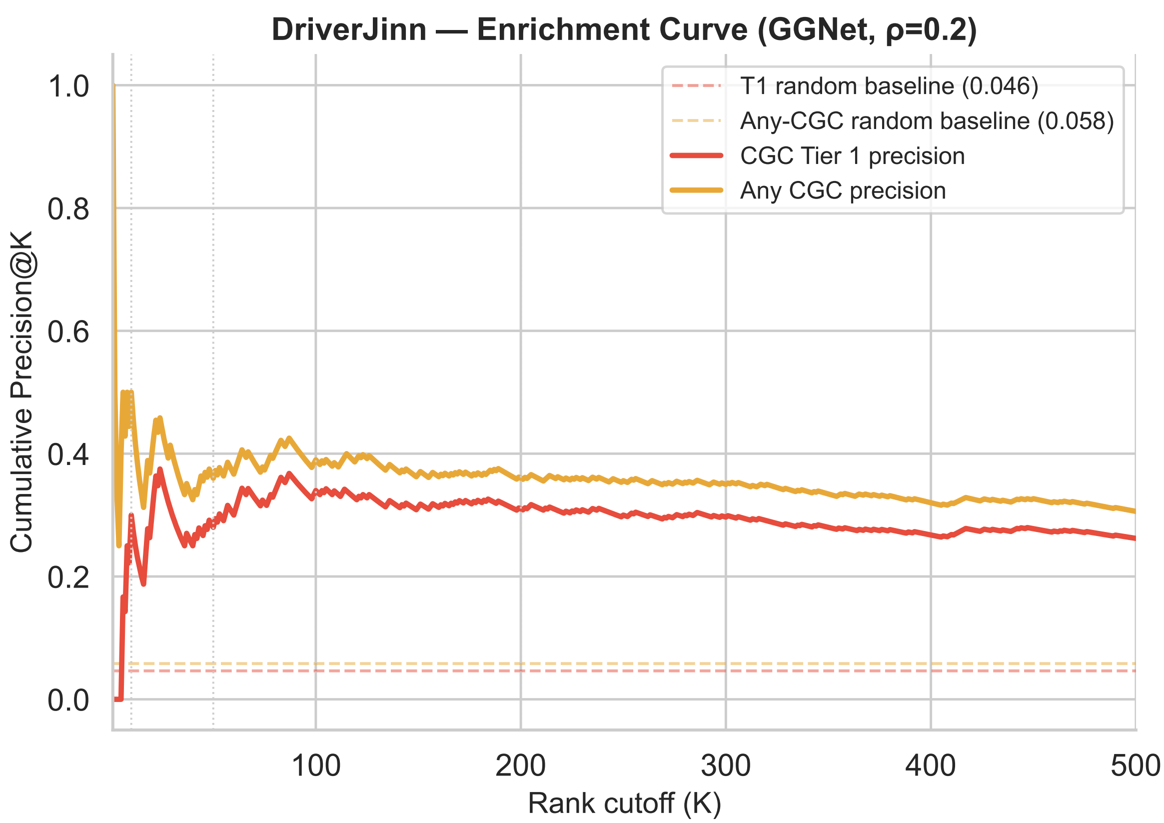
	Fold	Peak NDCG@50	Peak AUROC	Epochs
0	1	0.399200	0.753400	1080
1	2	0.502200	0.751900	680
2	3	0.426200	0.754200	780
3	4	0.524400	0.814400	1200
4	5	0.537500	0.766500	700
5	Mean	0.477900	0.768100	—

Enrichment Curve (Cumulative CGC Precision)

How quickly DriverJinn recovers known cancer driver genes as you move down the ranked list.

```
fig, ax = plt.subplots(figsize=(7, 5))
ks = np.arange(1, len(genes) + 1)
cumulative_t1  = np.cumsum(genes["cgc_tier"] == "T1")  / ks
cumulative_any = np.cumsum(genes["cgc_tier"] != "non-CGC") / ks
baseline_t1  = (genes["cgc_tier"] == "T1").sum()  / len(genes)
baseline_any = (genes["cgc_tier"] != "non-CGC").sum() / len(genes)

ax.axhline(baseline_t1,  color=COLORS["T1"], linestyle="--", linewidth=1.2,
           alpha=0.5, label=f"T1 random baseline ({baseline_t1:.3f})")
ax.axhline(baseline_any, color=COLORS["T2"], linestyle="--", linewidth=1.2,
           alpha=0.5, label=f"Any-CGC random baseline ({baseline_any:.3f})")
ax.plot(ks, cumulative_t1,  color=COLORS["T1"], linewidth=2.2, label="CGC Tier 1 precision")
ax.plot(ks, cumulative_any, color=COLORS["T2"], linewidth=2.2, label="Any CGC precision")
for k in [10, 50, 100, 200]:
    if k <= len(genes):
        ax.axvline(k, color="#CCCCCC", linewidth=0.8, linestyle=":")
ax.set_xlim(1, 500)
```

```
ax.set_xlabel("Rank cutoff (K)", fontsize=12)
ax.set_ylabel("Cumulative Precision@K", fontsize=12)
ax.set_title("DriverJinn - Enrichment Curve (GGNet,  =0.2)", fontsize=13, fontweight="bold")
ax.legend(fontsize=10)
sns.despine()
plt.tight_layout()
plt.savefig("poster_enrichment_curve.pdf", dpi=300, bbox_inches="tight")
plt.show()
```

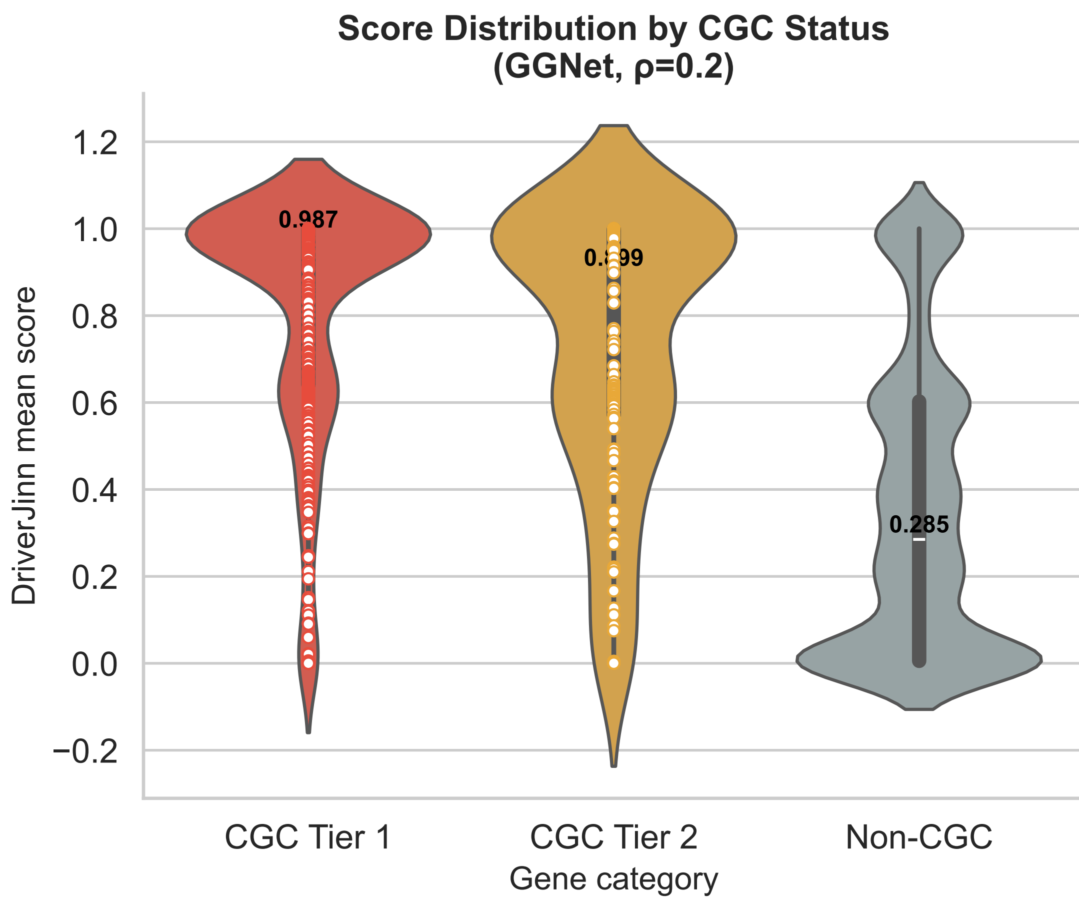


Score Distribution by CGC Tier

Separation of DriverJinn scores across CGC Tier 1, Tier 2, and non-CGC genes.

```python
fig, ax = plt.subplots(figsize=(6, 5))
order   = ["T1", "T2", "non-CGC"]
palette = {k: COLORS[k] for k in order}
sns.violinplot(data=genes, x="cgc_tier", y="mean_score",
               order=order, palette=palette, inner="box", linewidth=1.2, ax=ax)
for tier, color in [("T1", COLORS["T1"]), ("T2", COLORS["T2"])]:
    sub = genes[genes["cgc_tier"] == tier]
    ax.scatter([order.index(tier)] * len(sub), sub["mean_score"],
               color="white", edgecolors=color, s=18, zorder=5, linewidth=0.8)
for i, tier in enumerate(order):
    med = genes.loc[genes["cgc_tier"] == tier, "mean_score"].median()
    ax.text(i, med + 0.005, f"{med:.3f}", ha="center", va="bottom",
            fontsize=9, fontweight="bold", color="black")
ax.set_xlabel("Gene category", fontsize=12)
ax.set_ylabel("DriverJinn mean score", fontsize=12)
ax.set_title("Score Distribution by CGC Status\n(GGNet,  =0.2)", fontsize=13, fontweight="bol
ax.set_xticklabels(["CGC Tier 1", "CGC Tier 2", "Non-CGC"])
sns.despine()
plt.tight_layout()
plt.savefig("poster_score_violin.pdf", dpi=300, bbox_inches="tight")
plt.show()
```

```
C:\Users\siddh\AppData\Local\Temp\ipykernel_43672\372424057.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assig

  sns.violinplot(data=genes, x="cgc_tier", y="mean_score",
C:\Users\siddh\AppData\Local\Temp\ipykernel_43672\372424057.py:17: UserWarning: set_ticklabel
  ax.set_xticklabels(["CGC Tier 1", "CGC Tier 2", "Non-CGC"])
```

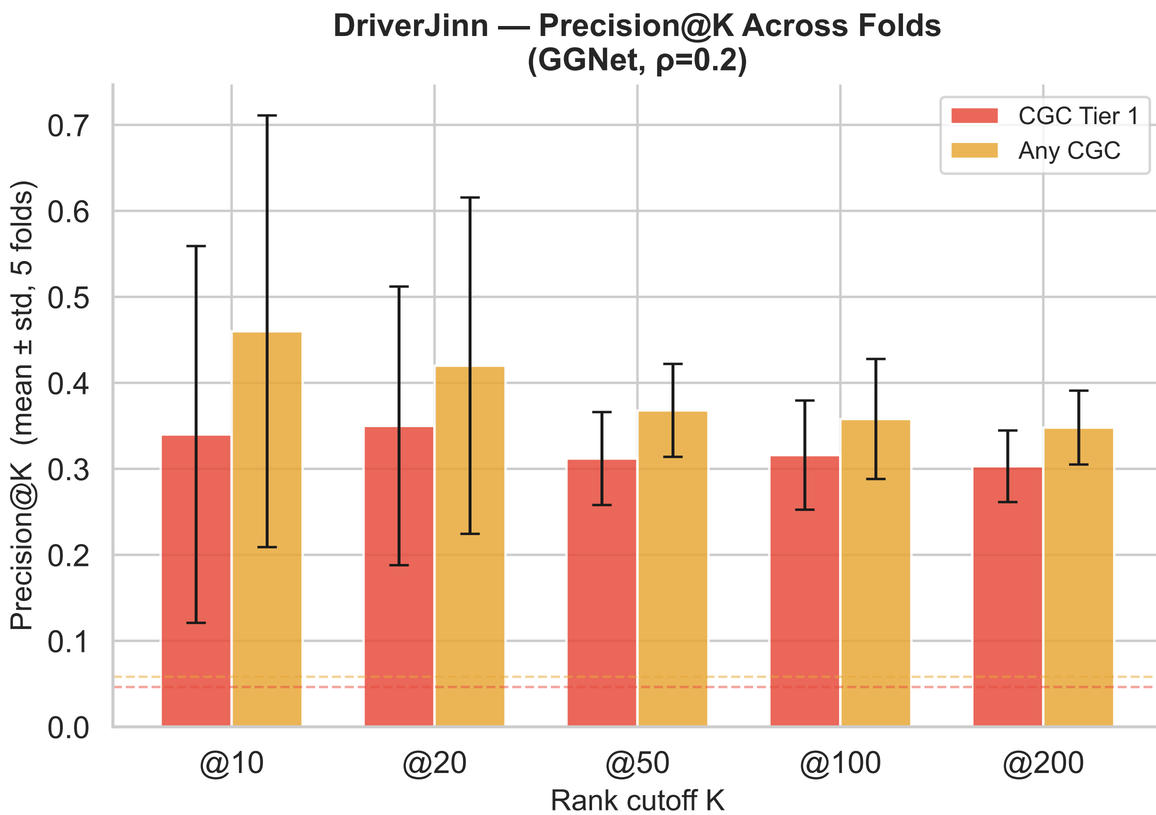


Precision@K Across Folds

Per-fold and mean Precision at multiple rank cutoffs, with error bars.

```
ks   = [10, 20, 50, 100, 200]
rows = []
for fold, df in fold_dfs.items():
    df = df.sort_values("rank").reset_index(drop=True)
    df["cgc_t1"]  = df["gene_name"].isin(cgc_t1)
    df["cgc_any"] = df["gene_name"].isin(cgc_t1 | cgc_t2)
    for k in ks:
        top_k = df.head(k)
        rows.append({"fold": fold, "K": k,
```

```
                     "Precision@K (T1)":      top_k["cgc_t1"].mean(),
                     "Precision@K (Any CGC)": top_k["cgc_any"].mean()})
prec_df = pd.DataFrame(rows)
summary = (prec_df.groupby("K")[["Precision@K (T1)", "Precision@K (Any CGC)"]]
           .agg(["mean", "std"]).reset_index())
summary.columns = ["K", "t1_mean", "t1_std", "any_mean", "any_std"]

fig, ax = plt.subplots(figsize=(7, 5))
width, xs = 0.35, np.arange(len(ks))
ax.bar(xs - width/2, summary["t1_mean"],  width, yerr=summary["t1_std"],
       color=COLORS["T1"], alpha=0.85, capsize=4, label="CGC Tier 1",
       error_kw={"linewidth": 1.2})
ax.bar(xs + width/2, summary["any_mean"], width, yerr=summary["any_std"],
       color=COLORS["T2"], alpha=0.85, capsize=4, label="Any CGC",
       error_kw={"linewidth": 1.2})
baseline_t1_v  = len(cgc_t1 & set(genes["gene_name"])) / len(genes)
baseline_any_v = len((cgc_t1 | cgc_t2) & set(genes["gene_name"])) / len(genes)
ax.axhline(baseline_t1_v,  color=COLORS["T1"], linestyle="--", linewidth=1, alpha=0.5)
ax.axhline(baseline_any_v, color=COLORS["T2"], linestyle="--", linewidth=1, alpha=0.5)
ax.set_xticks(xs)
ax.set_xticklabels([f"@{k}" for k in ks])
ax.set_xlabel("Rank cutoff K", fontsize=12)
ax.set_ylabel("Precision@K  (mean ± std, 5 folds)", fontsize=12)
ax.set_title("DriverJinn - Precision@K Across Folds\n(GGNet,  =0.2)", fontsize=13, fontweight
ax.legend(fontsize=10)
sns.despine()
plt.tight_layout()
plt.savefig("poster_precision_at_k.pdf", dpi=300, bbox_inches="tight")
plt.show()
```

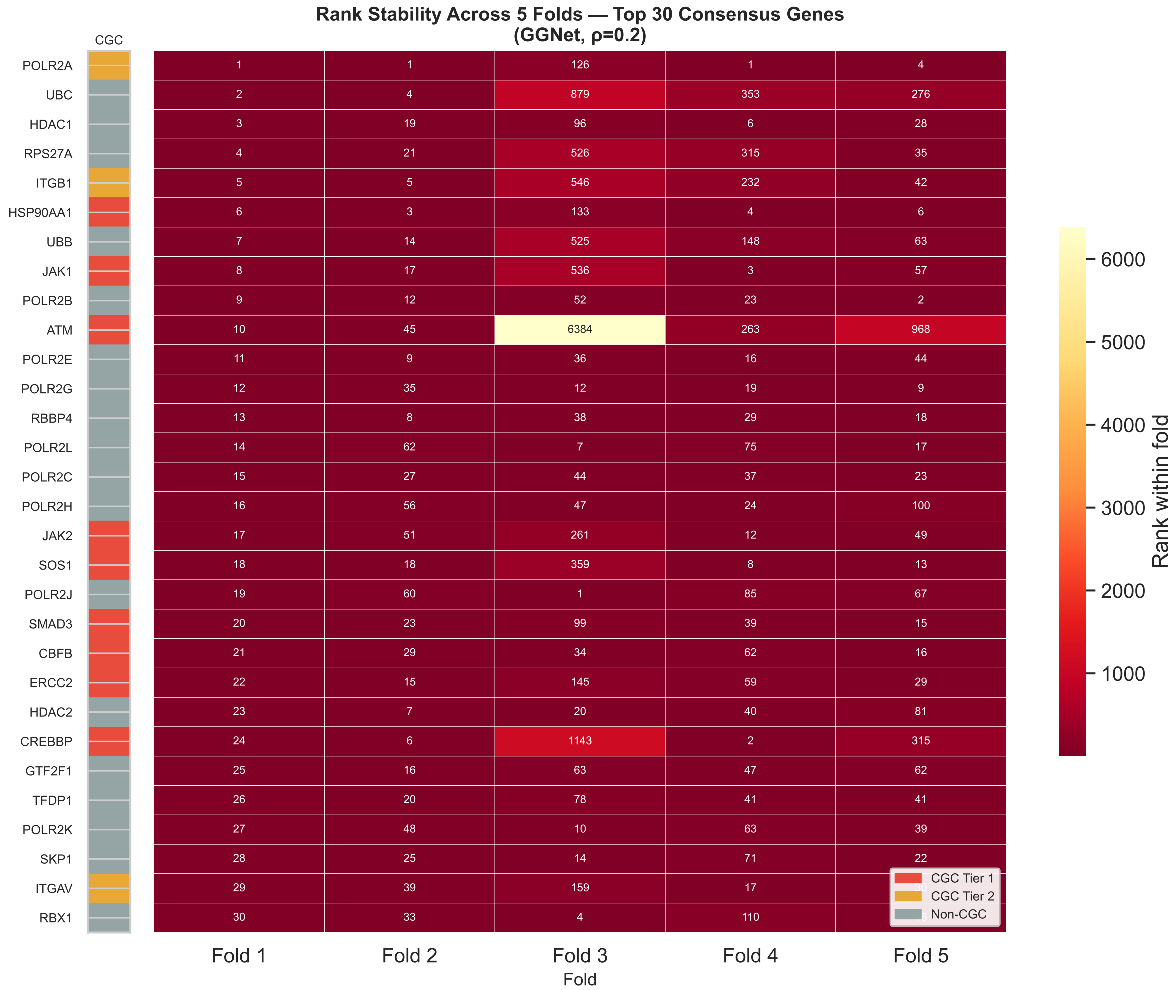


Rank Stability Heatmap

For the consensus top-30 genes, how consistent is their rank across all 5 folds?

```
TOP_N      = 30
top_genes = genes.head(TOP_N)["gene_name"].tolist()
rank_matrix = pd.DataFrame(index=top_genes)
for fold, df in fold_dfs.items():
    df = df.sort_values("rank")
    name_to_rank = dict(zip(df["gene_name"], df["rank"]))
    rank_matrix[f"Fold {fold}"] = [name_to_rank.get(g, np.nan) for g in top_genes]

cgc_array = np.array([[
    0 if g in cgc_t1 else 1 if g in cgc_t2 else 2
    for g in top_genes
```

```
]]).T

fig, axes = plt.subplots(1, 2, figsize=(11, 9),
                          gridspec_kw={"width_ratios": [0.04, 1]})
axes[0].imshow(cgc_array, aspect="auto",
               cmap=plt.cm.colors.ListedColormap([COLORS["T1"], COLORS["T2"], COLORS["non-CG
axes[0].set_xticks([])
axes[0].set_yticks(range(TOP_N))
axes[0].set_yticklabels(top_genes, fontsize=8)
axes[0].set_title("CGC", fontsize=8, pad=4)
sns.heatmap(rank_matrix, ax=axes[1], cmap="YlOrRd_r", annot=True, fmt=".0f",
            annot_kws={"size": 7}, linewidths=0.3,
            cbar_kws={"label": "Rank within fold", "shrink": 0.6})
axes[1].set_yticks([])
axes[1].set_xlabel("Fold", fontsize=11)
axes[1].set_title("Rank Stability Across 5 Folds - Top 30 Consensus Genes\n(GGNet,  =0.2)",
                   fontsize=12, fontweight="bold")
axes[1].legend(handles=[
    mpatches.Patch(color=COLORS["T1"],      label="CGC Tier 1"),
    mpatches.Patch(color=COLORS["T2"],      label="CGC Tier 2"),
    mpatches.Patch(color=COLORS["non-CGC"], label="Non-CGC"),
], loc="lower right", fontsize=8, framealpha=0.9)
plt.tight_layout()
plt.savefig("poster_rank_stability.pdf", dpi=300, bbox_inches="tight")
plt.show()
```

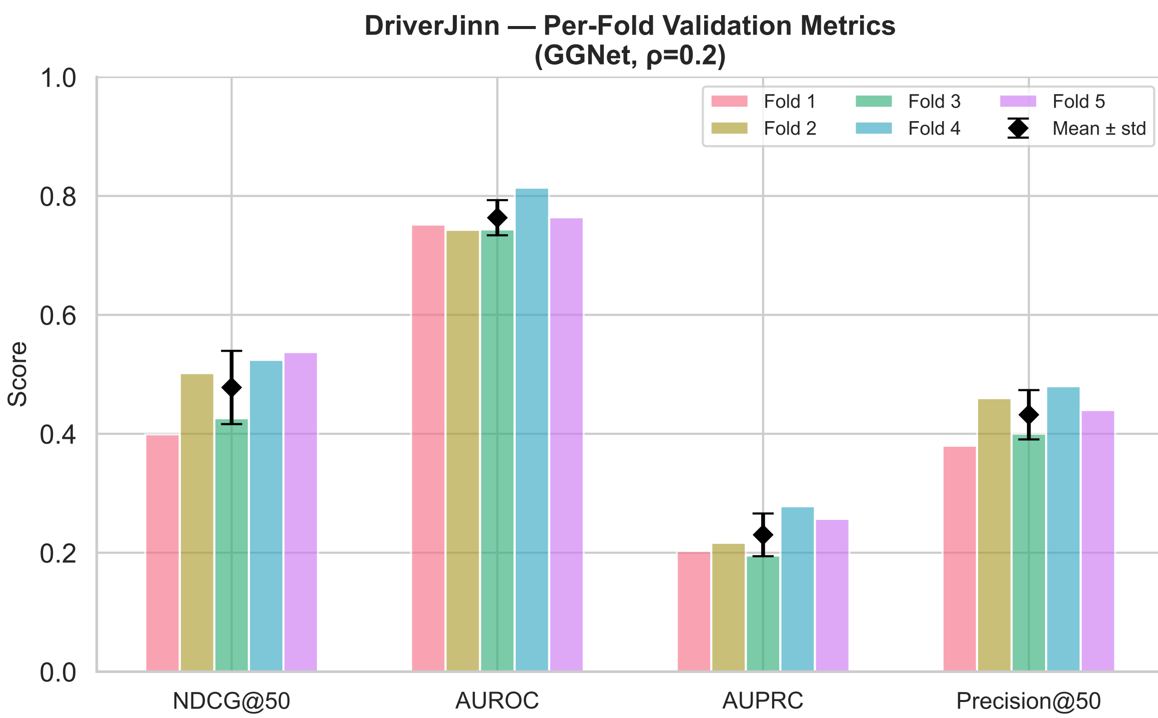


Per-Fold Metrics Summary

Overview of NDCG@50, AUROC, AUPRC, Precision@50 across all 5 folds.

```
metric_cols = ["NDCG@50", "AUROC", "AUPRC", "Precision@50"]
fold_data   = metrics_folds[["Fold"] + metric_cols].set_index("Fold")
mean_row    = metrics[metrics["Fold"] == "Mean"][metric_cols].values[0]
std_row     = metrics[metrics["Fold"] == "Std"][metric_cols].values[0]

fig, ax = plt.subplots(figsize=(8, 5))
```

```python
fold_palette = sns.color_palette("husl", 5)
xs    = np.arange(len(metric_cols))
width = 0.13
for i, (fold, row) in enumerate(fold_data.iterrows()):
    ax.bar(xs + (i - 2) * width, row[metric_cols],
           width, alpha=0.65, color=fold_palette[i], label=f"Fold {fold}")
ax.errorbar(xs, mean_row, yerr=std_row, fmt="D", color="black",
            markersize=6, linewidth=1.8, capsize=5, zorder=10, label="Mean ± std")
ax.set_xticks(xs)
ax.set_xticklabels(metric_cols, fontsize=11)
ax.set_ylabel("Score", fontsize=12)
ax.set_ylim(0, 1)
ax.set_title("DriverJinn - Per-Fold Validation Metrics\n(GGNet,  =0.2)",
             fontsize=13, fontweight="bold")
ax.legend(fontsize=9, ncol=3, loc="upper right")
sns.despine()
plt.tight_layout()
plt.savefig("poster_metrics_summary.pdf", dpi=300, bbox_inches="tight")
plt.show()
```



Hallmark Enrichment (ORA via Fisher's Exact Test)

```
def run_ora(gene_list, universe, hallmark_sets, min_gs_size=5, p_adj_method="fdr_bh"):
    gene_set = set(gene_list) & universe
    n_uni, n_query = len(universe), len(gene_set)
    rows = []
    for hallmark, gs in hallmark_sets.items():
        gs_in_uni = gs & universe
        if len(gs_in_uni) < min_gs_size:
            continue
        overlap_genes = gene_set & gs_in_uni
        a = len(overlap_genes)
        b = n_query - a
        c = len(gs_in_uni) - a
        d = n_uni - a - b - c
        _, pval = fisher_exact([[a, b], [c, d]], alternative="greater")
        rows.append({"Hallmark": hallmark, "Overlap": a,
                     "GS Size": len(gs_in_uni), "p-value": pval,
                     "Overlap Genes": ", ".join(sorted(overlap_genes))})
    if not rows:
        return pd.DataFrame()
    df = pd.DataFrame(rows)
    _, padj, _, _ = multipletests(df["p-value"], method=p_adj_method)
    df["p.adj"] = padj
    return df.sort_values("p-value").reset_index(drop=True)

hallmark_sets = (
    hallmarks_raw.dropna(subset=["HALLMARK"])
    .query("HALLMARK != ''")
    .groupby("HALLMARK")["GENE_SYMBOL"].apply(set).to_dict()
)
universe  = set(ranked_genes["gene_name"])
top100    = set(ranked_genes.head(100)["gene_name"])
consensus = set(ranked_genes.loc[ranked_genes["consensus_significant"], "gene_name"]) \
            if "consensus_significant" in ranked_genes.columns else top100
```

Top 100 Predicted Drivers

```
ora_top100 = run_ora(top100, universe, hallmark_sets)
if ora_top100.empty or (ora_top100["p.adj"] >= 0.05).all():
```

```
    print("No significant hallmark enrichment at p.adj < 0.05 for top 100")
else:
    sig = ora_top100[ora_top100["p.adj"] < 0.05].copy()
    sig["-log10(p.adj)"] = -np.log10(sig["p.adj"])
    fig, ax = plt.subplots(figsize=(7, max(3, len(sig) * 0.45)))
    sns.barplot(data=sig, x="-log10(p.adj)", y="Hallmark", palette="Blues_r", ax=ax)
    ax.axvline(-np.log10(0.05), color="red", linestyle="--", lw=1, label="p.adj = 0.05")
    ax.set_title("Hallmark Enrichment - Top 100 Predicted Drivers")
    ax.legend()
    plt.tight_layout()
    plt.show()
    display(sig[["Hallmark", "Overlap", "GS Size", "p-value", "p.adj"]]
            .style.format({"p-value": "{:.3e}", "p.adj": "{:.3e}"}))
```

```
C:\Users\siddh\AppData\Local\Temp\ipykernel_43672\2642632014.py:8: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assig

  sns.barplot(data=sig, x="-log10(p.adj)", y="Hallmark", palette="Blues_r", ax=ax)
```

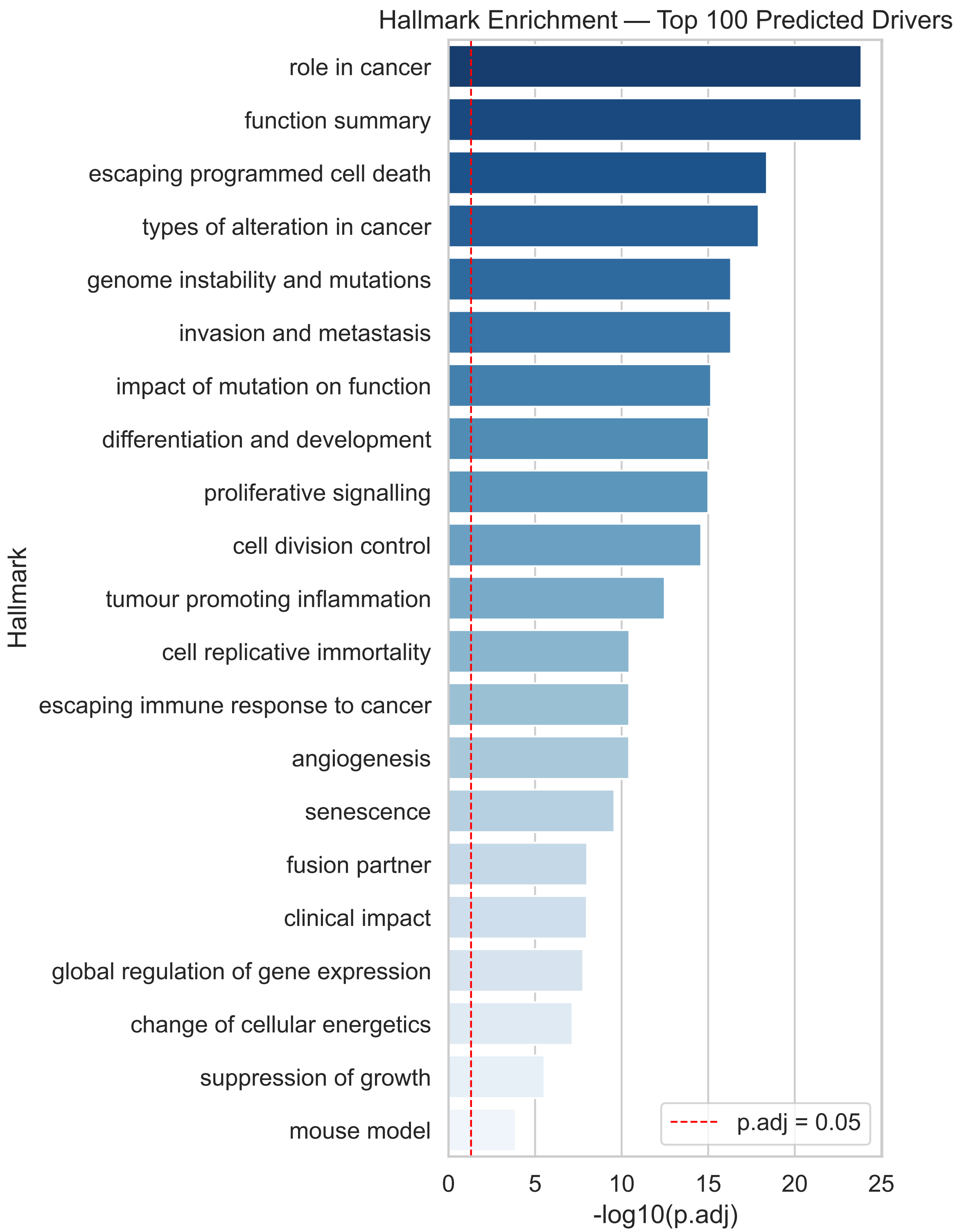

Hallmark Enrichment — Top 100 Predicted Drivers
role in cancer
function summary
escaping programmed cell death
types of alteration in cancer
genome instability and mutations
invasion and metastasis
impact of mutation on function
differentiation and development
proliferative signalling
cell division control
tumour promoting inflammation
cell replicative immortality
escaping immune response to cancer
angiogenesis
senescence
fusion partner
clinical impact
global regulation of gene expression
change of cellular energetics
suppression of growth
mouse model
Hallmark
p.adj = 0.05
0
5
10
15
20
25
-log10(p.adj)

Table 2

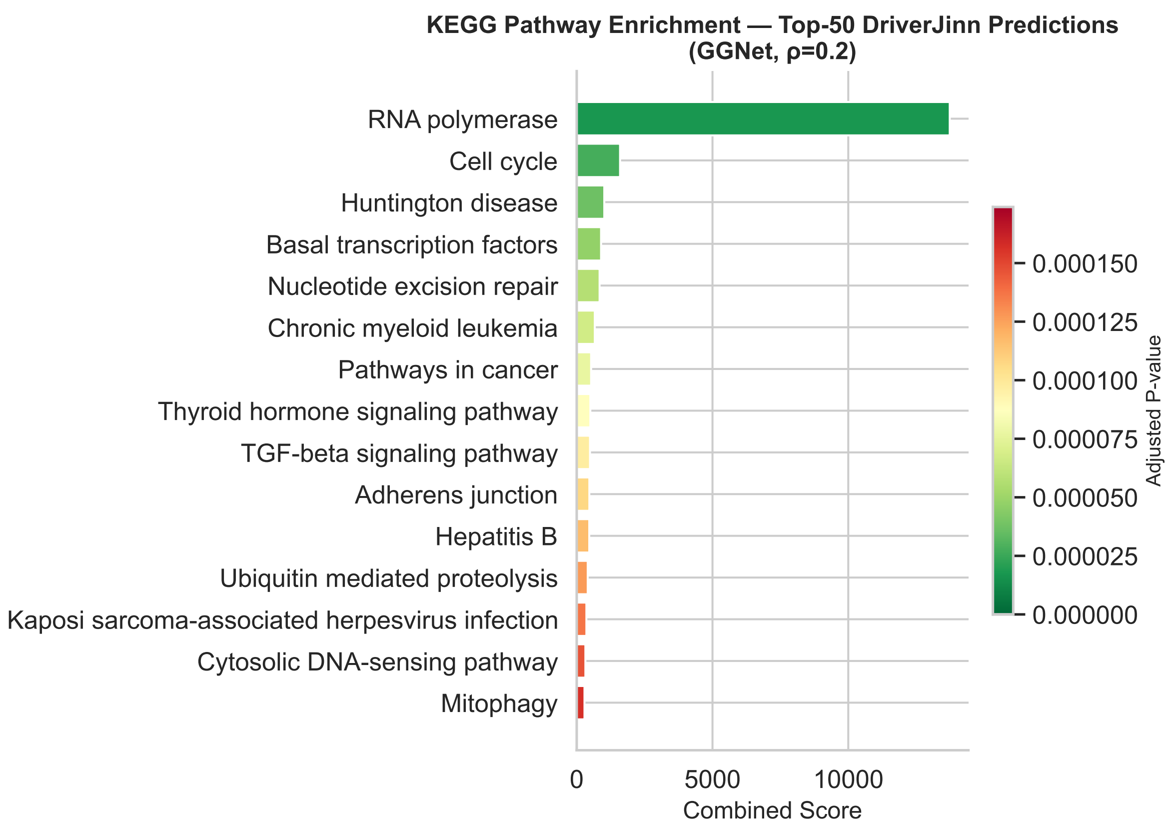
	Hallmark	Overlap	GS Size	p-value	p.adj
0	role in cancer	33	343	1.158e-25	1.401e-24
1	function summary	33	344	1.274e-25	1.401e-24
2	escaping programmed cell death	24	218	5.529e-20	4.055e-19
3	types of alteration in cancer	24	231	2.198e-19	1.209e-18
4	genome instability and mutations	18	119	1.211e-17	4.769e-17
5	invasion and metastasis	22	216	1.301e-17	4.769e-17
6	impact of mutation on function	20	189	2.170e-16	6.821e-16
7	differentiation and development	19	167	3.377e-16	9.285e-16
8	proliferative signalling	20	195	4.021e-16	9.830e-16
9	cell division control	17	129	1.148e-15	2.526e-15
10	tumour promoting inflammation	12	61	1.611e-13	3.223e-13
11	cell replicative immortality	10	51	1.949e-11	3.573e-11
12	escaping immune response to cancer	11	70	2.331e-11	3.682e-11
13	angiogenesis	12	91	2.343e-11	3.682e-11
14	senescence	10	63	1.785e-10	2.617e-10
15	fusion partner	12	148	7.134e-09	9.809e-09
16	clinical impact	11	119	7.924e-09	1.025e-08
17	global regulation of gene expression	9	72	1.355e-08	1.657e-08
18	change of cellular energetics	9	85	5.959e-08	6.900e-08
19	suppression of growth	9	132	2.598e-06	2.858e-06
20	mouse model	6	87	1.237e-04	1.296e-04

KEGG Pathway Enrichment

Requires `gseapy` (`pip install gseapy`) and internet access for Enrichr API.

```
try:
    import gseapy as gp
    top50_genes = genes.head(50)["gene_name"].tolist()
    enr = gp.enrichr(gene_list=top50_genes, gene_sets=["KEGG_2021_Human"],
                     organism="human", outdir=None, cutoff=0.05)
    results = (enr.results[enr.results["Adjusted P-value"] < 0.05]
               .sort_values("Combined Score", ascending=False).head(15))
    if results.empty:
        results = enr.results.sort_values("Adjusted P-value").head(15)
    fig, ax = plt.subplots(figsize=(8, 6))
```

```
    colors = plt.cm.RdYlGn_r(np.linspace(0.1, 0.9, len(results)))
    ax.barh(results["Term"].str.replace(r" \(.*\)", "", regex=True),
            results["Combined Score"], color=colors)
    sm = plt.cm.ScalarMappable(cmap="RdYlGn_r",
         norm=plt.Normalize(results["Adjusted P-value"].min(),
                            results["Adjusted P-value"].max()))
    sm.set_array([])
    plt.colorbar(sm, ax=ax, shrink=0.6).set_label("Adjusted P-value", fontsize=10)
    ax.set_xlabel("Combined Score", fontsize=12)
    ax.set_title("KEGG Pathway Enrichment - Top-50 DriverJinn Predictions\n(GGNet,  =0.2)",
                 fontsize=12, fontweight="bold")
    ax.invert_yaxis()
    sns.despine()
    plt.tight_layout()
    plt.savefig("poster_kegg_enrichment.pdf", dpi=300, bbox_inches="tight")
    plt.show()
except ImportError:
    print("gseapy not installed - run: pip install gseapy")
```



Novel Candidate Genes

Top consensus-significant predictions not in any CGC tier:

```
if "consensus_significant" in ranked_genes.columns:
    novel_df = (
        ranked_genes
        .loc[ranked_genes["consensus_significant"] & ~ranked_genes["cgc_any"]]
        [["aggregate_rank", "gene_name", "mean_score", "folds_significant", "total_folds"]]
        .head(20)
    )
    display(novel_df.style.format({"mean_score": "{:.4f}"}).hide(axis="index"))
else:
    print("consensus_significant column not found - skipping novel candidate table")
```

Table 3

aggregate_rank	gene_name	mean_score	folds_significant	total_folds
42	MNAT1	1.0000	5	5
59	TRAF6	0.9999	5	5
69	MRE11	0.9999	5	5
84	GTF2H1	0.9999	5	5
96	PDS5B	0.9999	5	5
106	DYNC1H1	0.9999	5	5
124	AGO3	0.9999	5	5
133	GTF2H4	0.9999	5	5
147	RPTOR	0.9999	5	5
174	SYNE1	0.9999	5	5
186	ARRB2	0.9999	5	5
188	LRP2	0.9999	5	5
210	YWHAB	0.9999	5	5
211	VCL	0.9999	5	5
215	PAK1	0.9999	5	5
216	MARK3	0.9999	5	5
221	BTRC	0.9999	5	5
239	REV3L	0.9999	5	5
251	SNRPB	0.9999	5	5
257	VWF	0.9999	5	5

Cross-Model Consensus

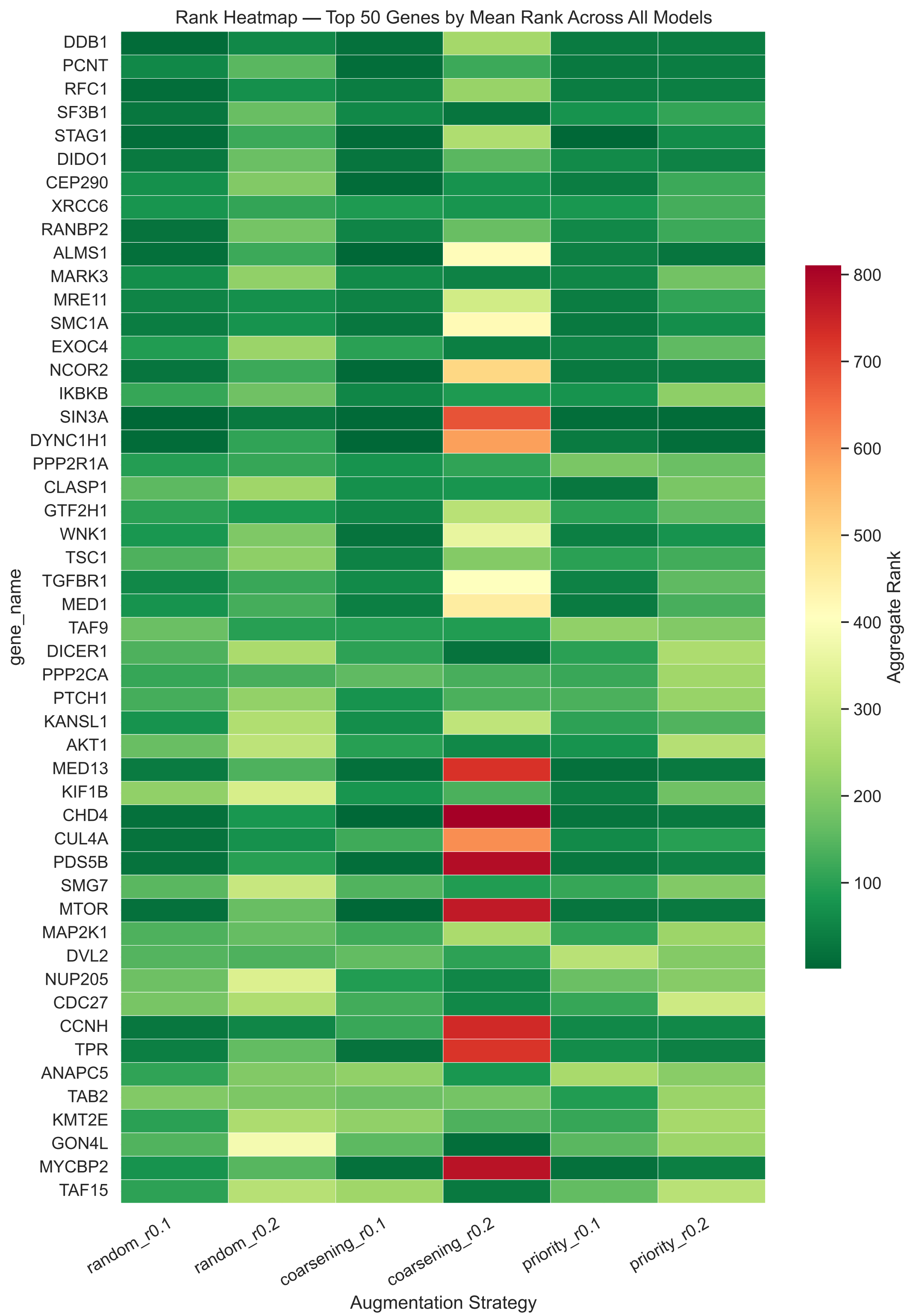
Combining ranked lists from 6 augmentation strategies to identify genes consistently predicted across strategies.

```
MODEL_LISTS = {
    "random_r0.1"    : "../model_results_2026-03-14/GGNet_random_r0.1_aggregated/GGNet_randon
    "random_r0.2"    : "../model_results_2026-03-14/GGNet_random_r0.2_aggregated/GGNet_randon
    "coarsening_r0.1": "../model_results_2026-03-15/GGNet_coarsening_r0.1_aggregated/GGNet_co
    "coarsening_r0.2": "../model_results_2026-03-15/GGNet_coarsening_r0.2_aggregated/GGNet_co
    "priority_r0.1"  : "../model_results_2026-03-15/GGNet_priority_r0.1_aggregated/GGNet_pric
    "priority_r0.2"  : "../model_results_2026-03-15/GGNet_priority_r0.2_aggregated/GGNet_pric
}

dfs = {}
for model, path in MODEL_LISTS.items():
    df = pd.read_csv(path)[["gene_name", "aggregate_rank", "mean_score", "consensus_significa
    df = df.rename(columns={"aggregate_rank": f"rank_{model}",
                             "mean_score": f"score_{model}",
                             "consensus_significant": f"consensus_{model}"})
    dfs[model] = df

merged_models = reduce(lambda l, r: pd.merge(l, r, on="gene_name", how="outer"), dfs.values()

rank_cols  = [f"rank_{m}"      for m in MODEL_LISTS]
score_cols = [f"score_{m}"     for m in MODEL_LISTS]
cons_cols  = [f"consensus_{m}" for m in MODEL_LISTS]

merged_models["mean_rank"]        = merged_models[rank_cols].mean(axis=1)
merged_models["mean_score_all"]   = merged_models[score_cols].mean(axis=1)
merged_models["models_consensus"] = merged_models[cons_cols].sum(axis=1).astype(int)
merged_models["cgc_tier"] = merged_models["gene_name"].map(
    lambda g: "Tier 1" if g in cgc_tier1 else ("Tier 2" if g in cgc_tier2 else "Novel")
)
cross_consensus = merged_models.sort_values("mean_rank").reset_index(drop=True)

print(f"Total genes across all models : {len(cross_consensus)}")
print(f"Consensus in  3/6 models      : {(cross_consensus['models_consensus'] >= 3).sum()}")
print(f"Consensus in  4/6 models      : {(cross_consensus['models_consensus'] >= 4).sum()}")
print(f"Consensus in all 6 models     : {(cross_consensus['models_consensus'] == 6).sum()}")
```

```
Total genes across all models : 11183
Consensus in  3/6 models      : 74
Consensus in  4/6 models      : 6
Consensus in all 6 models      : 0
```

Rank Heatmap — Top 50 by Mean Rank

```
top_n   = 50
top_ids = cross_consensus.head(top_n)["gene_name"].tolist()
heat_df = cross_consensus.set_index("gene_name").loc[top_ids, rank_cols].copy()
heat_df.columns = list(MODEL_LISTS.keys())
fig, ax = plt.subplots(figsize=(10, max(6, top_n * 0.28)))
sns.heatmap(heat_df, cmap="RdYlGn_r", ax=ax, linewidths=0.3, linecolor="white",
            cbar_kws={"label": "Aggregate Rank", "shrink": 0.6})
ax.set_xlabel("Augmentation Strategy")
ax.set_title(f"Rank Heatmap - Top {top_n} Genes by Mean Rank Across All Models")
ax.set_xticklabels(ax.get_xticklabels(), rotation=30, ha="right")
plt.tight_layout()
plt.show()
```


Rank Heatmap — Top 50 Genes by Mean Rank Across All Models
DDB1
PCNT
RFC1
SF3B1
STAG1
DIDO1
CEP290
XRCC6
RANBP2
ALMS1
MARK3
MRE11
SMC1A
EXOC4
NCOR2
IKBKB
SIN3A
DYNC1H1
PPP2R1A
CLASP1
GTF2H1
WNK1
TSC1
TGFBR1
MED1
TAF9
DICER1
PPP2CA
PTCH1
KANSL1
AKT1
MED13
KIF1B
CHD4
CUL4A
PDS5B
SMG7
MTOR
MAP2K1
DVL2
NUP205
CDC27
CCNH
TPR
ANAPC5
TAB2
KMT2E
GON4L
MYCBP2
TAF15
gene_name
random_r0.1
random_r0.2
coarsening_r0.1
coarsening_r0.2
priority_r0.1
priority_r0.2
Augmentation Strategy
800
700
600
500
400
300
200
100
Aggregate Rank

Top Cross-Model Consensus Genes (4/6 models)

```
cols_display = ["gene_name", "mean_rank", "mean_score_all", "models_consensus", "cgc_tier"] +
fmt = {"mean_rank": "{:.1f}", "mean_score_all": "{:.4f}"}
fmt.update({c: "{:.0f}" for c in rank_cols})
strong_4 = cross_consensus[cross_consensus["models_consensus"] >= 4][cols_display].reset_inde
strong_4.index += 1
strong_4.style.format(fmt).background_gradient(subset=rank_cols, cmap="RdYlGn_r", axis=None)
```

Table

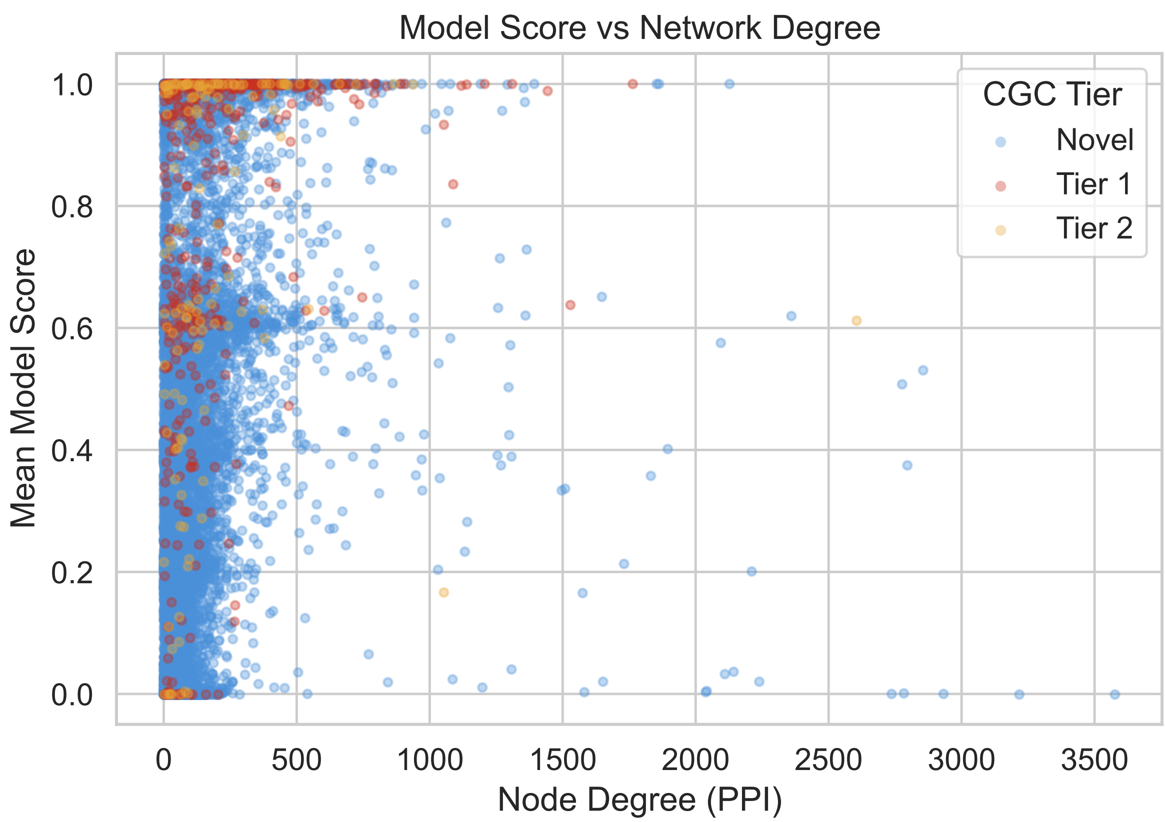
	gene_name	mean_rank	mean_score_all	models_consensus	cgc_tier	rank_random_r0.1	rank_r
1	TRERF1	439.5	0.6662	4	Novel	240	436
2	CBFA2T2	489.3	0.6862	4	Novel	441	693
3	ANAPC2	998.0	0.6053	4	Novel	1322	798
4	NPAS2	1112.3	0.5251	4	Novel	974	1190
5	SLC16A1	1419.7	0.4960	4	Novel	1569	1545
6	TUB	4395.7	0.2400	4	Novel	2837	4021

PPI Network Analysis

Degree Centrality vs Model Score

```
ranked_genes["degree"] = ranked_genes["gene_name"].map(
    lambda g: degree_map.get(name_to_idx.get(g), np.nan)
)

fig, ax = plt.subplots(figsize=(7, 5))
for tier, grp in ranked_genes.dropna(subset=["degree"]).groupby("cgc_label"):
    ax.scatter(grp["degree"], grp["mean_score"], alpha=0.35, s=12,
               color=tier_colors[tier], label=tier, rasterized=True)
ax.set_xlabel("Node Degree (PPI)")
ax.set_ylabel("Mean Model Score")
ax.set_title("Model Score vs Network Degree")
ax.legend(title="CGC Tier")
plt.tight_layout()
```

```
plt.show()

valid = ranked_genes.dropna(subset=["degree"])
rho, pval = spearmanr(valid["degree"], valid["mean_score"])
print(f"Spearman rho (degree vs score): {rho:.3f}  p={pval:.2e}")
```

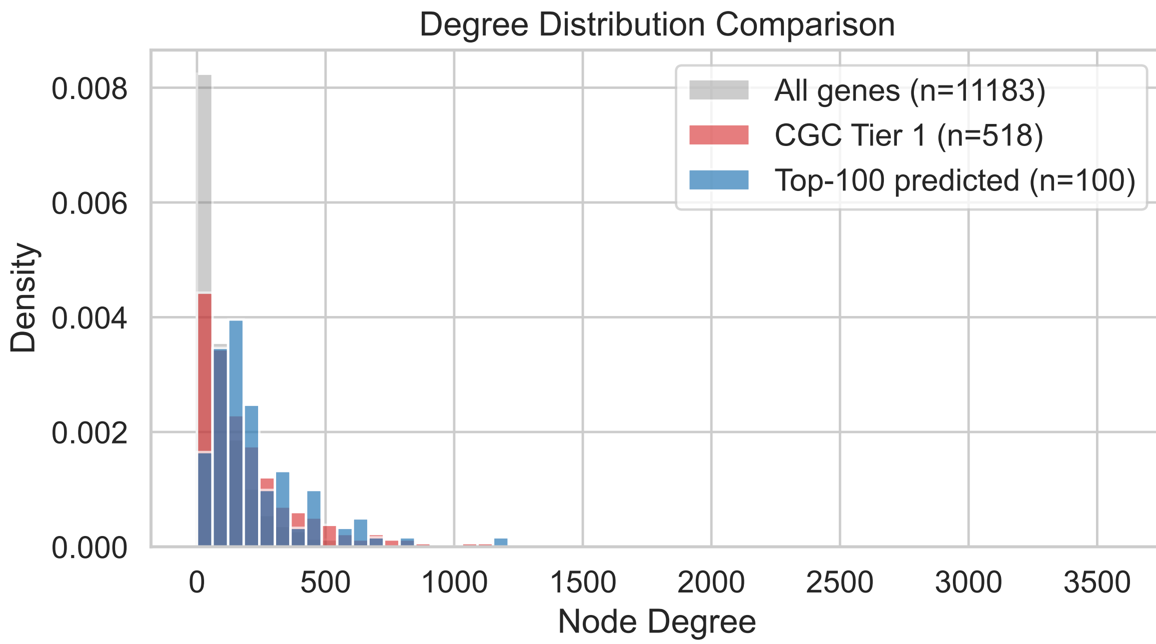



```
Spearman rho (degree vs score): 0.476  p=0.00e+00
```

Degree Distribution: Top-100 vs CGC vs All Genes

```
all_degrees    = [degree_map.get(name_to_idx.get(g), 0) for g in ranked_genes["gene_name"] i
top100_degrees = [degree_map.get(name_to_idx.get(g), 0) for g in ranked_genes.head(100)["gene
cgc1_degrees   = [degree_map.get(name_to_idx.get(g), 0) for g in ranked_genes[ranked_genes["c

fig, ax = plt.subplots(figsize=(7, 4))
```

```
bins = np.linspace(0, max(all_degrees), 60)
ax.hist(all_degrees,    bins=bins, alpha=0.4, label=f"All genes (n={len(all_degrees)})",
ax.hist(cgc1_degrees,   bins=bins, alpha=0.6, label=f"CGC Tier 1 (n={len(cgc1_degrees)})",
ax.hist(top100_degrees, bins=bins, alpha=0.7, label=f"Top-100 predicted (n={len(top100_degree
ax.set_xlabel("Node Degree")
ax.set_ylabel("Density")
ax.set_title("Degree Distribution Comparison")
ax.legend()
plt.tight_layout()
plt.show()
print(f"Median degree - All: {np.median(all_degrees):.0f} | "
      f"CGC Tier 1: {np.median(cgc1_degrees):.0f} | "
      f"Top-100: {np.median(top100_degrees):.0f}")
```

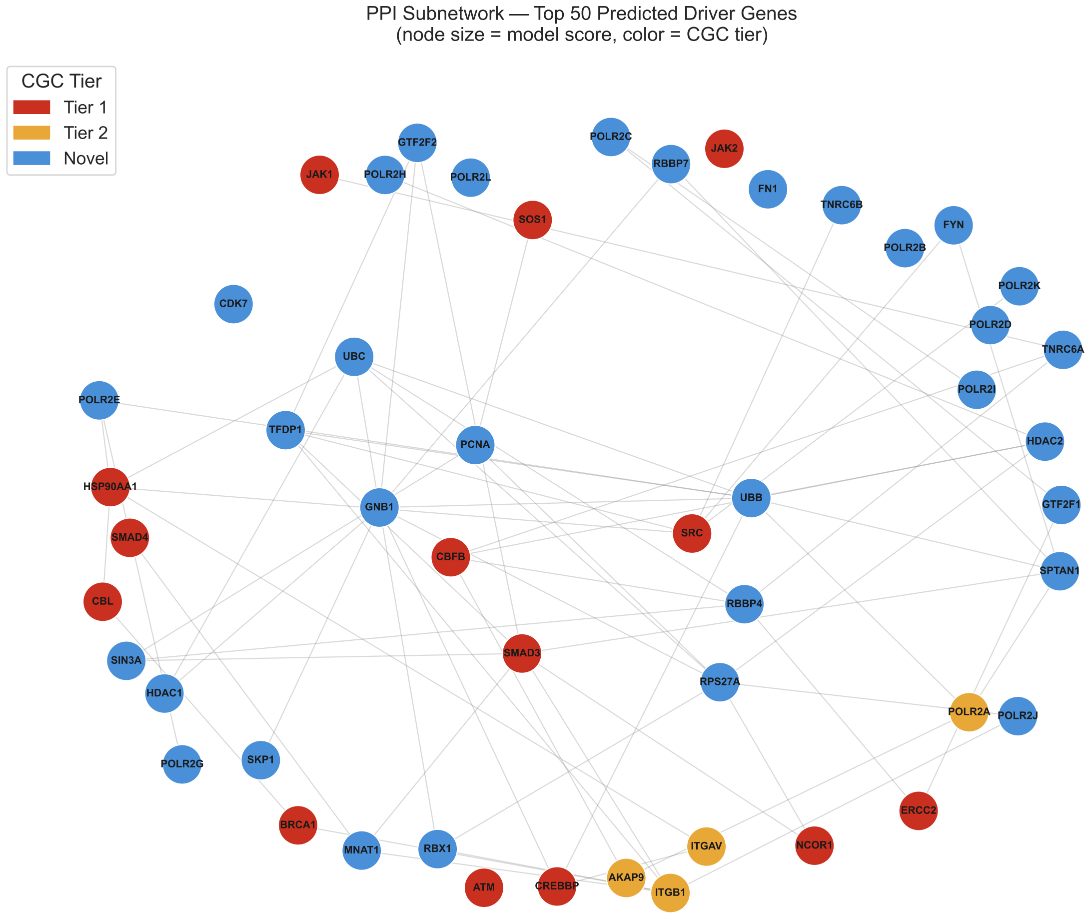



```
Median degree - All: 61 | CGC Tier 1: 126 | Top-100: 177
```

PPI Subnetwork of Top-50 Predicted Genes

```
top50_nodes = [name_to_idx[g] for g in genes.head(50)["gene_name"] if g in name_to_idx]
G_sub = G_full.subgraph(top50_nodes).copy()
```

```
G_named = nx.relabel_nodes(G_sub, id_to_name)

score_map_50  = genes.set_index("gene_name")["mean_score"].to_dict()
tier_map_50   = genes.set_index("gene_name")["cgc_label"].to_dict()
node_colors50 = [tier_colors.get(tier_map_50.get(n, "Novel"), "#1f77b4") for n in G_named.noc
node_sizes50  = [score_map_50.get(n, 0.5) * 800 for n in G_named.nodes()]

pos = nx.spring_layout(G_named, seed=42, k=1.5)
fig, ax = plt.subplots(figsize=(12, 10))
nx.draw_networkx_edges(G_named, pos, ax=ax, alpha=0.3, edge_color="grey", width=0.8)
nx.draw_networkx_nodes(G_named, pos, ax=ax, node_color=node_colors50,
                       node_size=node_sizes50, edgecolors="white", linewidths=0.8)
nx.draw_networkx_labels(G_named, pos, ax=ax, font_size=7.5, font_weight="bold")
ax.legend(handles=[Patch(color=c, label=t) for t, c in tier_colors.items()],
          title="CGC Tier", loc="upper left")
ax.set_title("PPI Subnetwork - Top 50 Predicted Driver Genes\n"
             "(node size = model score, color = CGC tier)", pad=12)
ax.axis("off")
plt.tight_layout()
plt.show()
print(f"Subgraph density: {nx.density(G_named):.4f}")
connected = list(nx.connected_components(G_named))
print(f"Connected components: {len(connected)} (largest: {max(len(c) for c in connected)} noc
```




```
Subgraph density: 0.0555
Connected components: 8 (largest: 43 nodes)
```

Cross-Network Consensus Table

Top 20 genes ranked consistently across GGNet, PathNet, and PPNet.

```
ggnet   = pd.read_csv("../model_results_2026-03-14/GGNet_random_r0.2_aggregated/"
                      "GGNet_random_r0.2_aggregated_all_genes.csv")
pathnet = pd.read_csv("../model_results_2026-03-20/PathNet_random_r0.2_aggregated/"
                      "PathNet_random_r0.2_aggregated_all_genes.csv")
```

```python
ppnet   = pd.read_csv("../model_results_2026-03-20/PPNet_random_r0.2_aggregated/"
                      "PPNet_random_r0.2_aggregated_all_genes.csv")

def cgc_label(gene):
    if gene in cgc_tier1: return "Tier 1"
    if gene in cgc_tier2: return "Tier 2"
    return "Novel"

merged_net = (
    ggnet[["gene_name", "aggregate_rank", "mean_score"]]
    .rename(columns={"aggregate_rank": "GGNet_rank", "mean_score": "GGNet_score"})
    .merge(pathnet[["gene_name", "aggregate_rank", "mean_score"]]
           .rename(columns={"aggregate_rank": "PathNet_rank", "mean_score": "PathNet_score"}
           on="gene_name", how="inner")
    .merge(ppnet[["gene_name", "aggregate_rank", "mean_score"]]
           .rename(columns={"aggregate_rank": "PPNet_rank", "mean_score": "PPNet_score"}),
           on="gene_name", how="inner")
)
merged_net["CGC"] = merged_net["gene_name"].apply(cgc_label)
merged_net["consensus_rank"] = merged_net[["GGNet_rank", "PathNet_rank", "PPNet_rank"]].mean
merged_net = merged_net.sort_values("consensus_rank")

top20 = merged_net.head(20).copy()
top20.insert(0, "Consensus Rank", range(1, 21))
top20 = top20.rename(columns={"gene_name": "Gene", "GGNet_rank": "GGNet",
                               "PathNet_rank": "PathNet", "PPNet_rank": "PPNet",
                               "CGC": "CGC Status"})
display_cols = ["Consensus Rank", "Gene", "GGNet", "PathNet", "PPNet", "CGC Status"]

ggnet_scores = ggnet.set_index("gene_name")["mean_score"].to_dict()

def style_row(row):
    if row["CGC Status"] == "Novel":  return ["background-color: #FFF3CD; font-weight: bold"
    if row["CGC Status"] == "Tier 1": return ["background-color: #FDECEA"] * len(row)
    return [""] * len(row)

styled = (
    top20[display_cols].style
    .apply(style_row, axis=1)
    .format({"GGNet": "{:.0f}", "PathNet": "{:.0f}", "PPNet": "{:.0f}"})
    .set_caption("Top 20 Consensus Cancer Driver Gene Predictions Across Networks "
                 "(DriverJinn,  =0.2) - Novel candidates highlighted in gold")
```

```
    .set_table_styles([
        {"selector": "caption", "props": [("font-size","13px"),("font-weight","bold"),
                                          ("text-align","left"),("padding-bottom","8px")]},
        {"selector": "th",      "props": [("background-color","#2C3E50"),("color","white"),
                                          ("font-size","11px"),("text-align","center"),
                                          ("padding","6px 10px")]},
        {"selector": "td",      "props": [("font-size","11px"),("text-align","center"),
                                          ("padding","5px 10px")]},
        {"selector": "tr:hover td", "props": [("background-color","#EBF5FB")]},
    ])
)

novel_top20 = top20[top20["CGC Status"] == "Novel"]["Gene"].tolist()
print(f"Novel candidates in top 20: {novel_top20}")
display(styled)

import dataframe_image as dffi
dffi.export(styled, "novel.png", table_conversion="matplotlib", dpi=200)
```

```
Novel candidates in top 20: ['SIN3A', 'HDAC2', 'SMARCC1', 'KMT2E', 'SMARCC2', 'SMARCA2', 'RBI
```

Table 5: Top 20 Consensus Cancer Driver Gene Predictions Across Networks (DriverJinn, =0.2) — Novel candidates highlighted in gold

	Consensus Rank	Gene	GGNet	PathNet	PPNet	CGC Status
30	1	SIN3A	31	20	118	Novel
22	2	HDAC2	23	82	69	Novel
113	3	NCOR2	121	46	91	Tier 1
55	4	SMARCC1	58	149	85	Novel
224	5	KMT2E	255	22	23	Novel
17	6	SOS1	18	91	193	Tier 1
20	7	CBFB	21	84	209	Tier 1
93	8	KDR	100	113	103	Tier 1
85	9	SMARCC2	92	29	196	Novel
171	10	SMARCA2	190	68	61	Novel
12	11	RBBP4	13	47	294	Novel
196	12	EED	218	34	115	Tier 2
82	13	FRS2	88	132	147	Novel
254	14	DOCK7	290	37	64	Novel
280	15	NCOA3	319	4	80	Novel

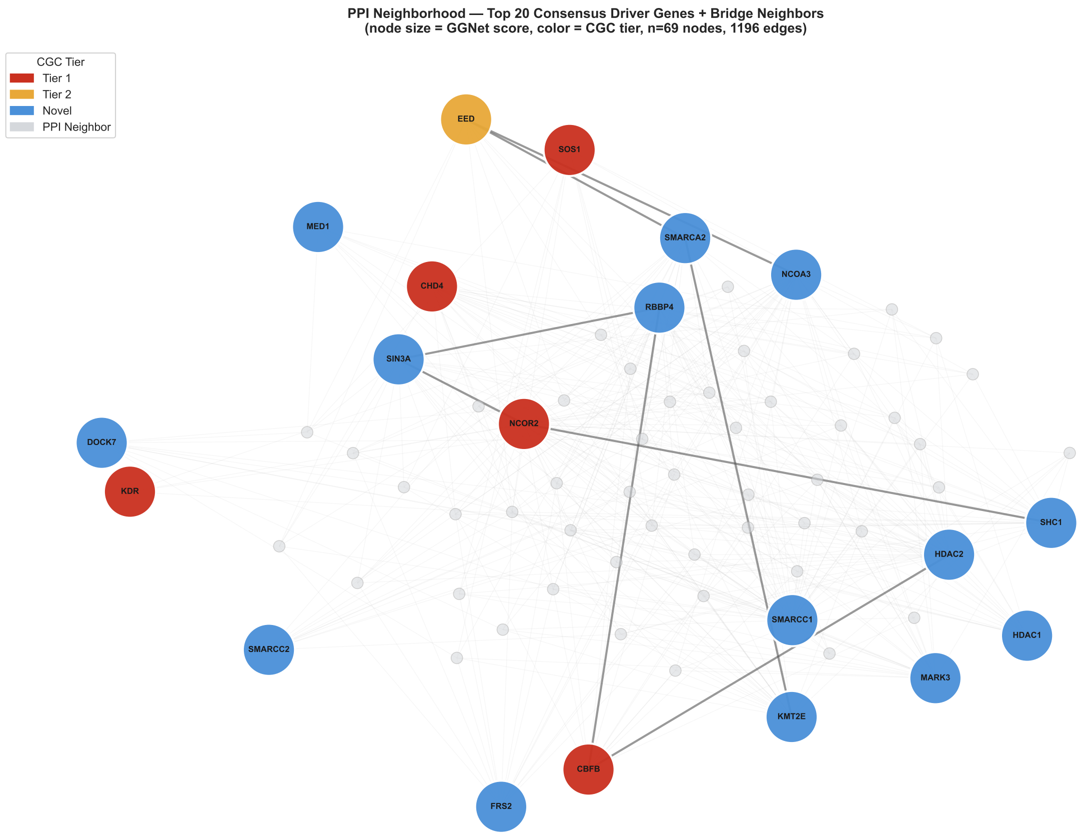
	Consensus Rank	Gene	GGNet	PathNet	PPNet	CGC Status
120	16	MED1	129	108	170	Novel
76	17	CHD4	82	19	315	Tier 1
95	18	SHC1	102	281	36	Novel
195	19	MARK3	216	107	100	Novel
2	20	HDAC1	3	122	301	Novel

PPI Neighborhood Subgraph (Consensus Top 20)

Top 20 consensus genes + bridge neighbors connecting 6 of them.

```
target_genes = top20["Gene"].tolist()
target_ids   = [name_to_id[g] for g in target_genes if g in name_to_id]
found_genes  = [g for g in target_genes if g in name_to_id]
target_id_set = set(target_ids)

# 1-hop neighbors connecting  6 target genes
neighbor_target_count = {}
for nid in target_ids:
    for nb in G_full.neighbors(nid):
        if nb not in target_id_set:
            neighbor_target_count[nb] = neighbor_target_count.get(nb, 0) + 1
filtered_neighbors = {nb for nb, cnt in neighbor_target_count.items() if cnt >= 6}
neighbor_ids = target_id_set | filtered_neighbors

H_raw = G_full.subgraph(neighbor_ids).copy()
H     = nx.relabel_nodes(H_raw, id_to_name)

target_set    = set(found_genes)
cgc_color_map = {"Tier 1": "#ca301f", "Tier 2": "#E8A838", "Novel": "#4A90D9"}
cgc_lookup    = top20.set_index("Gene")["CGC Status"].to_dict()

# Edges
target_edges  = [(u, v) for u, v in H.edges() if u in target_set and v in target_set]
spoke_edges   = [(u, v) for u, v in H.edges() if (u in target_set) != (v in target_set)]

fig, ax = plt.subplots(figsize=(16, 12))
pos = nx.spring_layout(H, seed=42, k=1.5)
```

```
nx.draw_networkx_edges(H, pos, edgelist=spoke_edges,  ax=ax, alpha=0.10, edge_color="#AAAAAA
nx.draw_networkx_edges(H, pos, edgelist=target_edges, ax=ax, alpha=0.60, edge_color="#555555

neighbor_nodes = [g for g in H.nodes() if g not in target_set]
nx.draw_networkx_nodes(H, pos, nodelist=neighbor_nodes, ax=ax,
                       node_color="#D5D8DC", node_size=120,
                       edgecolors="#AAAAAA", linewidths=0.8, alpha=0.55)
target_in_H = list(target_set & set(H.nodes()))
nx.draw_networkx_nodes(H, pos, nodelist=target_in_H, ax=ax,
                       node_color=[cgc_color_map.get(cgc_lookup.get(g, "Novel"), "#4A90D9")
                                   for g in target_in_H],
                       node_size=[max(ggnet_scores.get(g, 0.5), 0.1) * 2500 for g in target_
                       edgecolors="white", linewidths=2.0, alpha=0.95)
nx.draw_networkx_labels(H, pos, labels={g: g for g in target_set if g in H.nodes()},
                        ax=ax, font_size=8, font_weight="bold", font_color="#1A1A1A")

patches = [mpatches.Patch(color=c, label=l) for l, c in cgc_color_map.items()]
patches.append(mpatches.Patch(color="#D5D8DC", label="PPI Neighbor"))
ax.legend(handles=patches, title="CGC Tier", fontsize=11, title_fontsize=11,
          loc="upper left", framealpha=0.9)
ax.set_title(f"PPI Neighborhood - Top 20 Consensus Driver Genes + Bridge Neighbors\n"
             f"(node size = GGNet score, color = CGC tier, "
             f"n={H.number_of_nodes()} nodes, {H.number_of_edges()} edges)",
             fontsize=13, fontweight="bold", pad=14)
ax.axis("off")
plt.tight_layout()
plt.savefig("consensus_subnetwork.pdf", dpi=300, bbox_inches="tight")
plt.savefig("consensus_subnetwork.png", dpi=200, bbox_inches="tight")
plt.show()
print(f"Target genes in graph : {sorted(target_set & set(H.nodes()))}")
print(f"Total nodes           : {H.number_of_nodes()}")
print(f"Total edges           : {H.number_of_edges()}")
```




```
Target genes in graph : ['CBFB', 'CHD4', 'DOCK7', 'EED', 'FRS2', 'HDAC1', 'HDAC2', 'KDR', 'KI
Total nodes           : 69
Total edges           : 1196
```